

Nepali FinTech Solutions

Automating Financial Data Extraction

OIM 7502: Advanced Programming Business Analytics

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Why Financial Data Extraction Matters in Nepal

A fragmented public market with no structured financial data



Nepal's Market Reality

- Public companies publish only PDFs (no APIs, no database, no central archive)
- No historical datasets for analysis or transparency
- Investors rely on rumors and sentiment, not fundamentals

PDF & Language Challenges

- PDFs range from 10–500 pages with inconsistent layouts
- Mixed Nepali + English (e.g., “ब्याजआय”)
- Fiscal years follow Bikram Sambat (e.g., 2080/81)
- No standard reporting format

Why an AI Pipeline Is Needed

- Nepal has no Bloomberg/FactSet-style tools
- No backtesting or financial terminals
- AI is the only scalable way to classify reports, extract fields, and create structured datasets

Inspiration

Tables and UI/UX

The logo for Yahoo! Finance, featuring the word "yahoo!" in a bold, dark blue font with an exclamation mark, and the word "finance" in a smaller, lighter blue font below it.

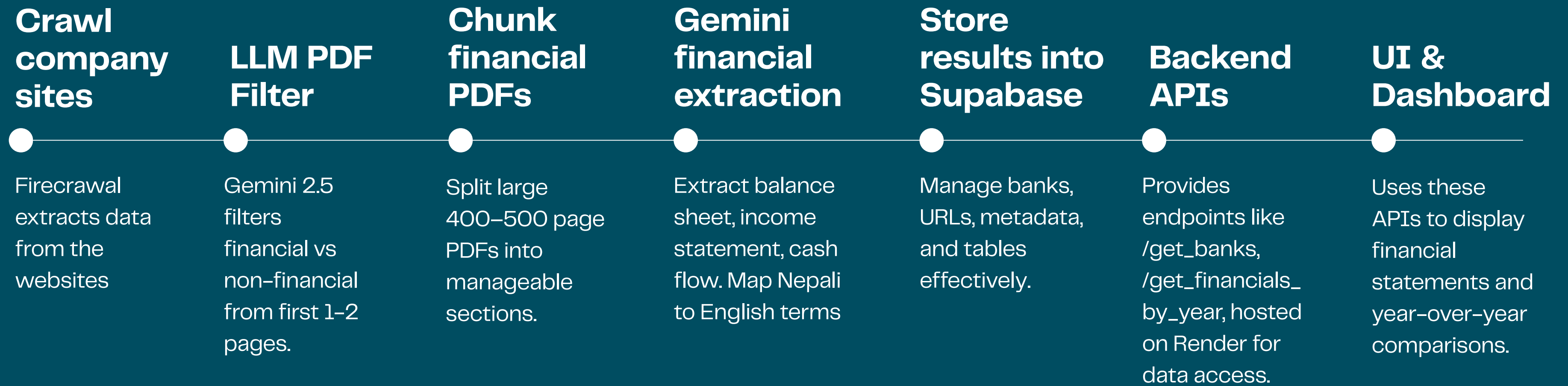
Analytics and APIs

The logo for FactSet, featuring the word "FACTSET" in a bold, blue, sans-serif font.

Algorithms and Back testing

The logo for Bloomberg, featuring the word "Bloomberg" in a bold, white, sans-serif font, centered within a solid blue rectangular background.

Workflow & Architecture

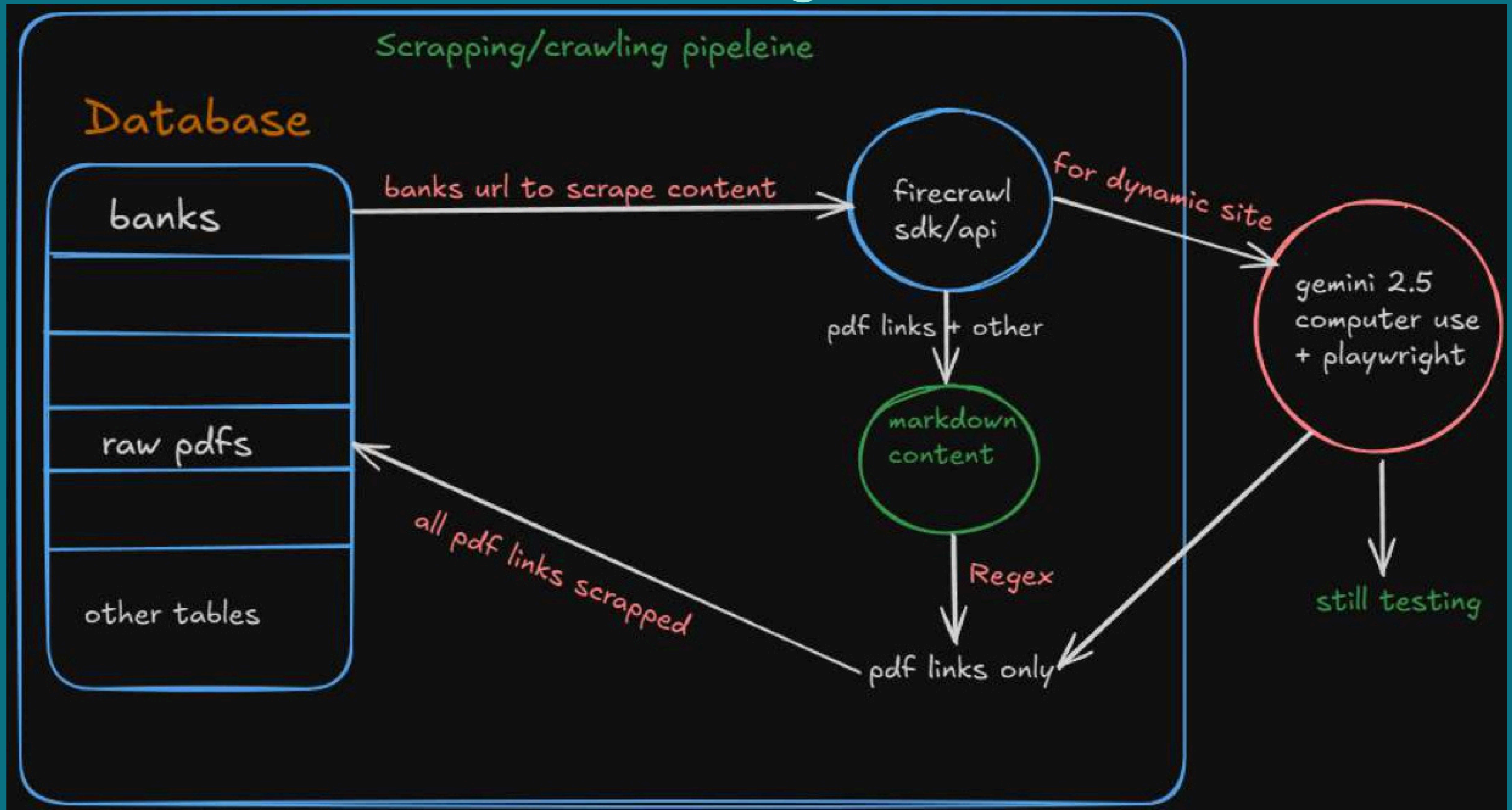


Tools and Technologies Evaluated

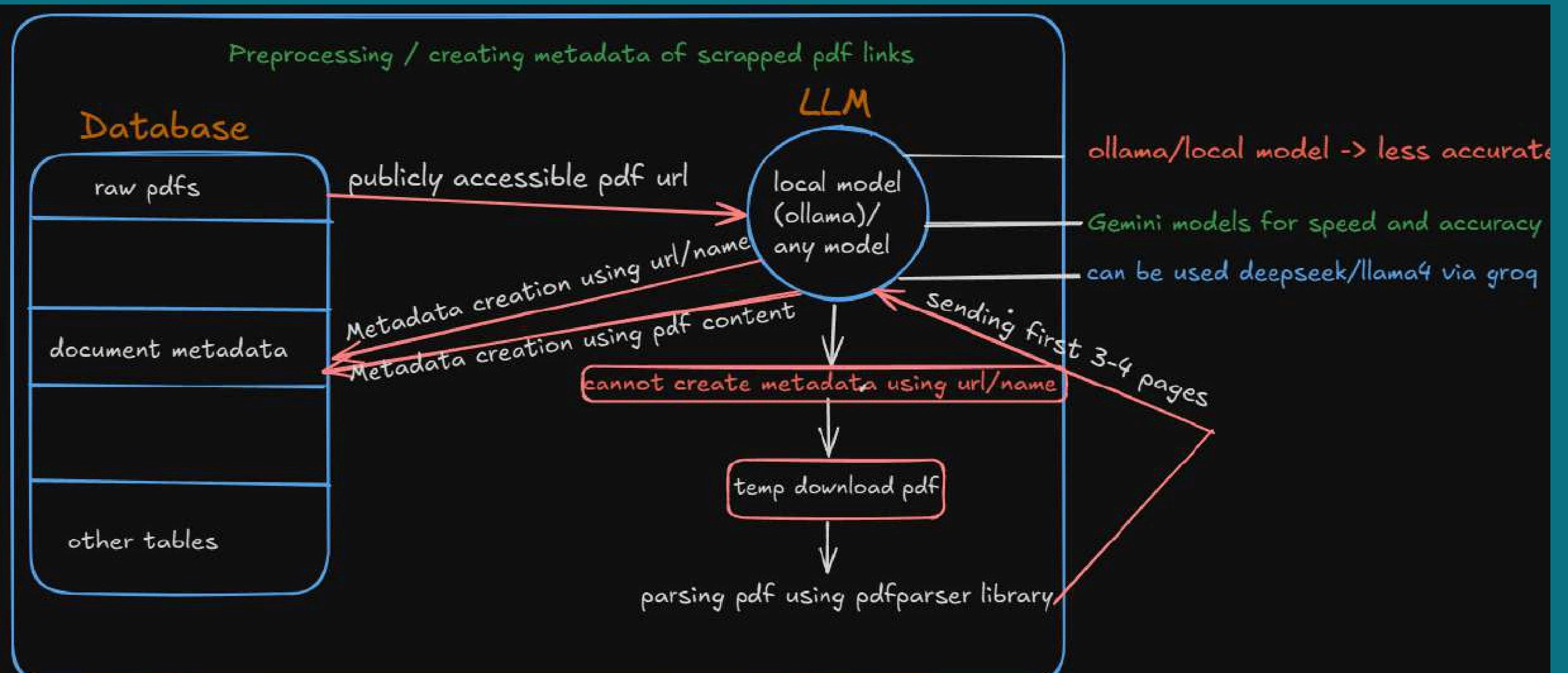


TOOL	OBSERVATION	OUTCOME
BeautifulSoup / Requests / Scrapy	Effective only for simple HTML structures. Failed for JavaScript-rendered sites and modern dynamic pages.	Discarded for production use.
Selenium / Playwright	Simulates browser actions effectively. However, slow and resource-intensive for multi-domain scraping.	Used only for fully dynamic sites.
Firecrawl	Excellent for static and semi-dynamic content. Supports advanced scraping with scrape() and extract() methods, and integrates with AI agents.	Adopted as the core scraping framework
Gemini 2.5 Computer Use	Tested for intelligent automation and UI interaction. However, no free tier is available and usage costs are high.	Not selected for continuous scraping due to high operational cost.

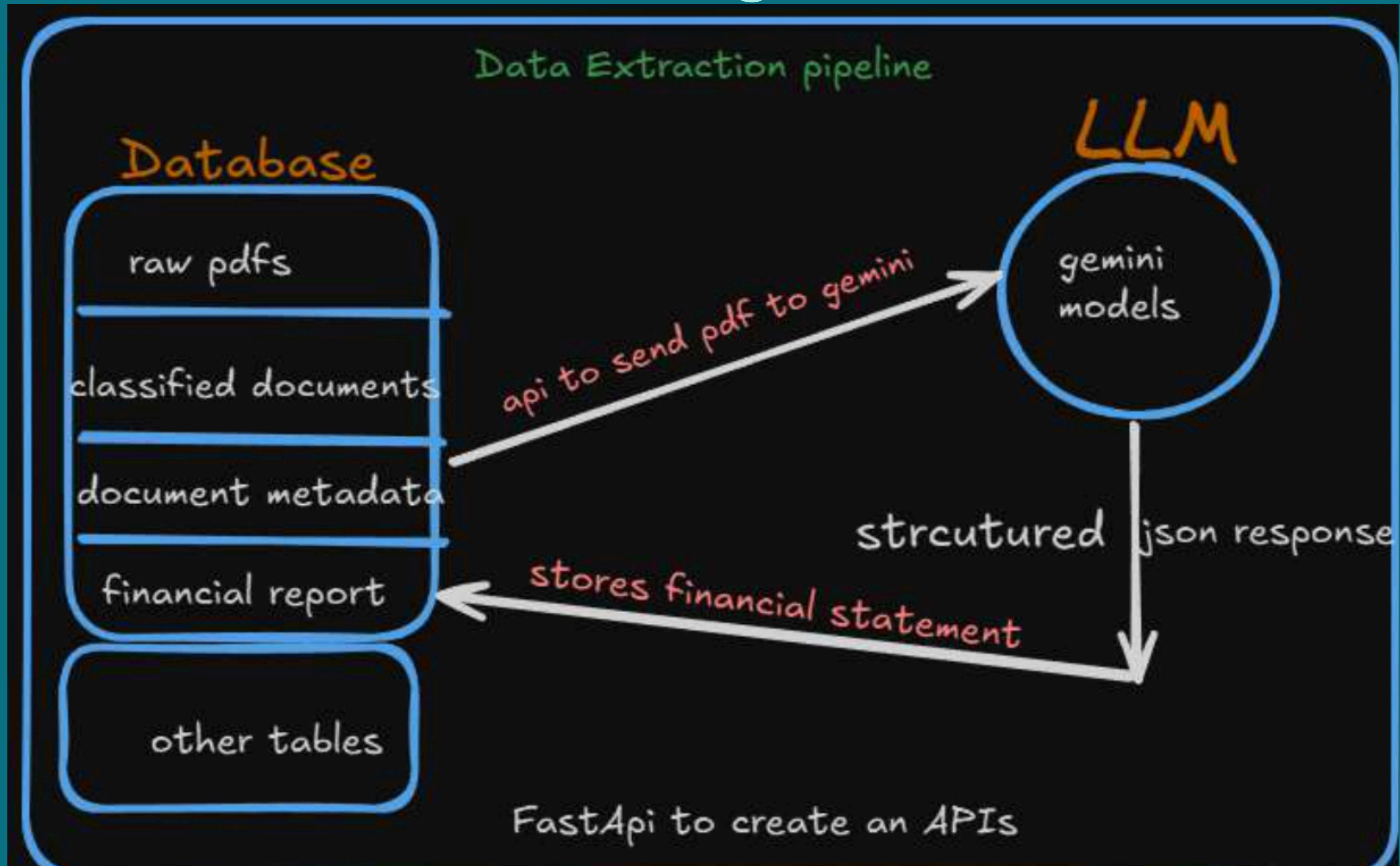
Architecture and Flow diagram



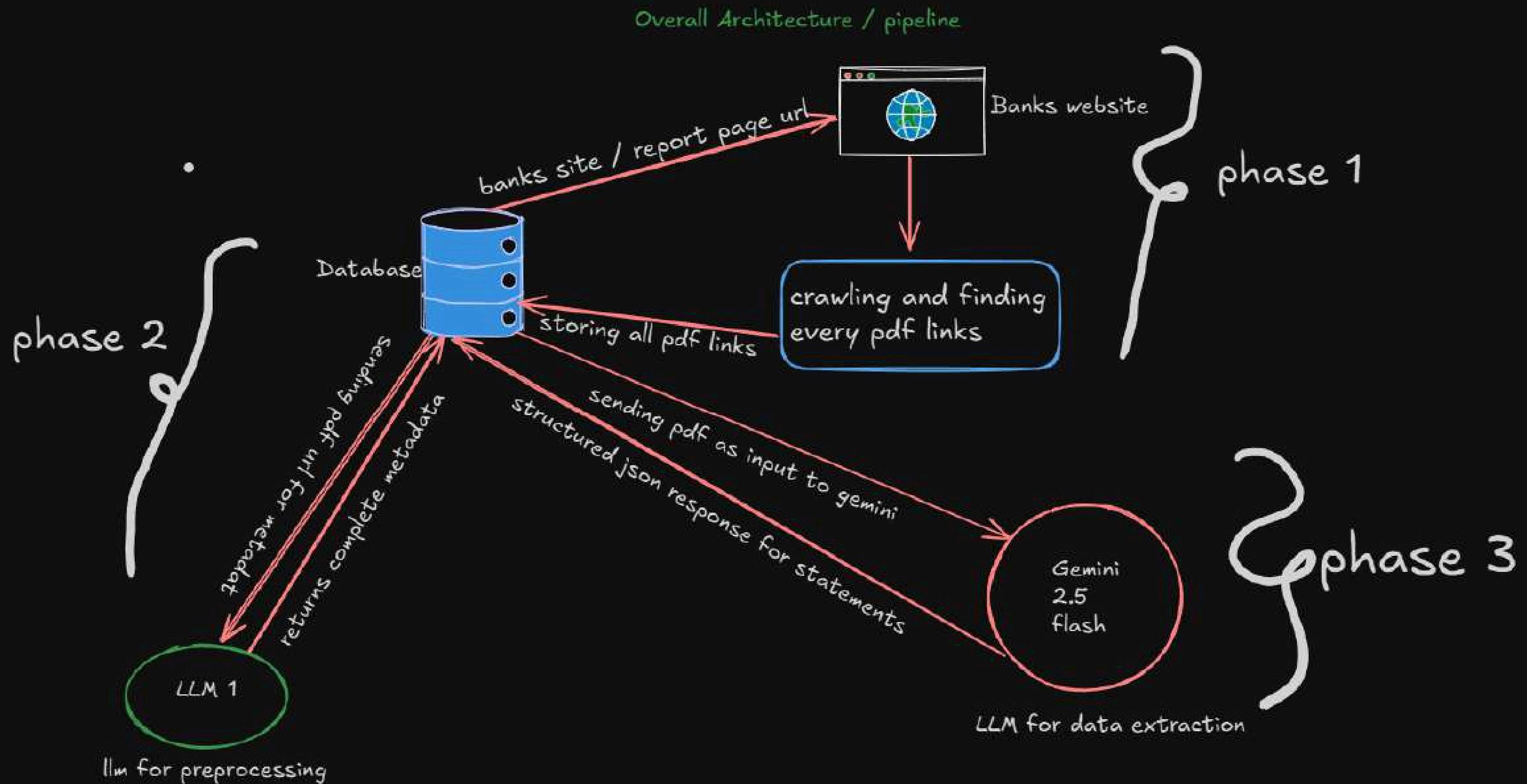
Architecture and Flow diagram



Architecture and Flow diagram



Architecture and Flow diagram



Python Script Overview

Modular structure for crawling, filtering, extraction, and storage

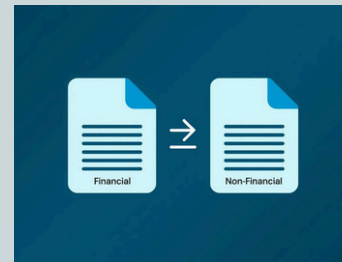
scraper

Crawls company websites and collects PDF links from static and dynamic pages.



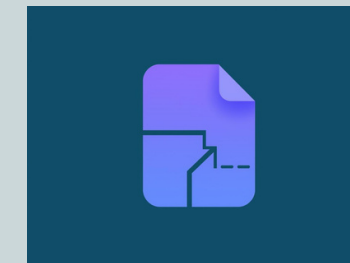
classifier

Uses the first 1–2 pages to classify PDFs, detect report type, and identify fiscal year.



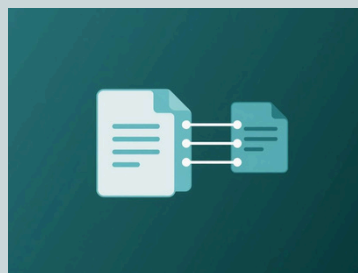
chunker

Splits large 400–500 page PDFs into manageable LLM-ready chunks.



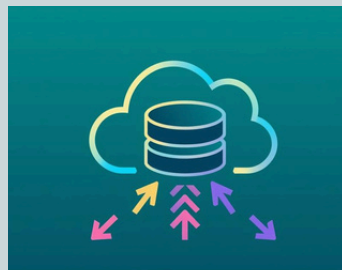
extractor

Extracts balance sheet, income statement, and cash flow tables; maps Nepali → English.



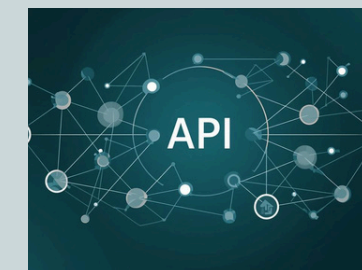
database

Stores URLs, metadata, classifications, and extracted tables in Supabase.

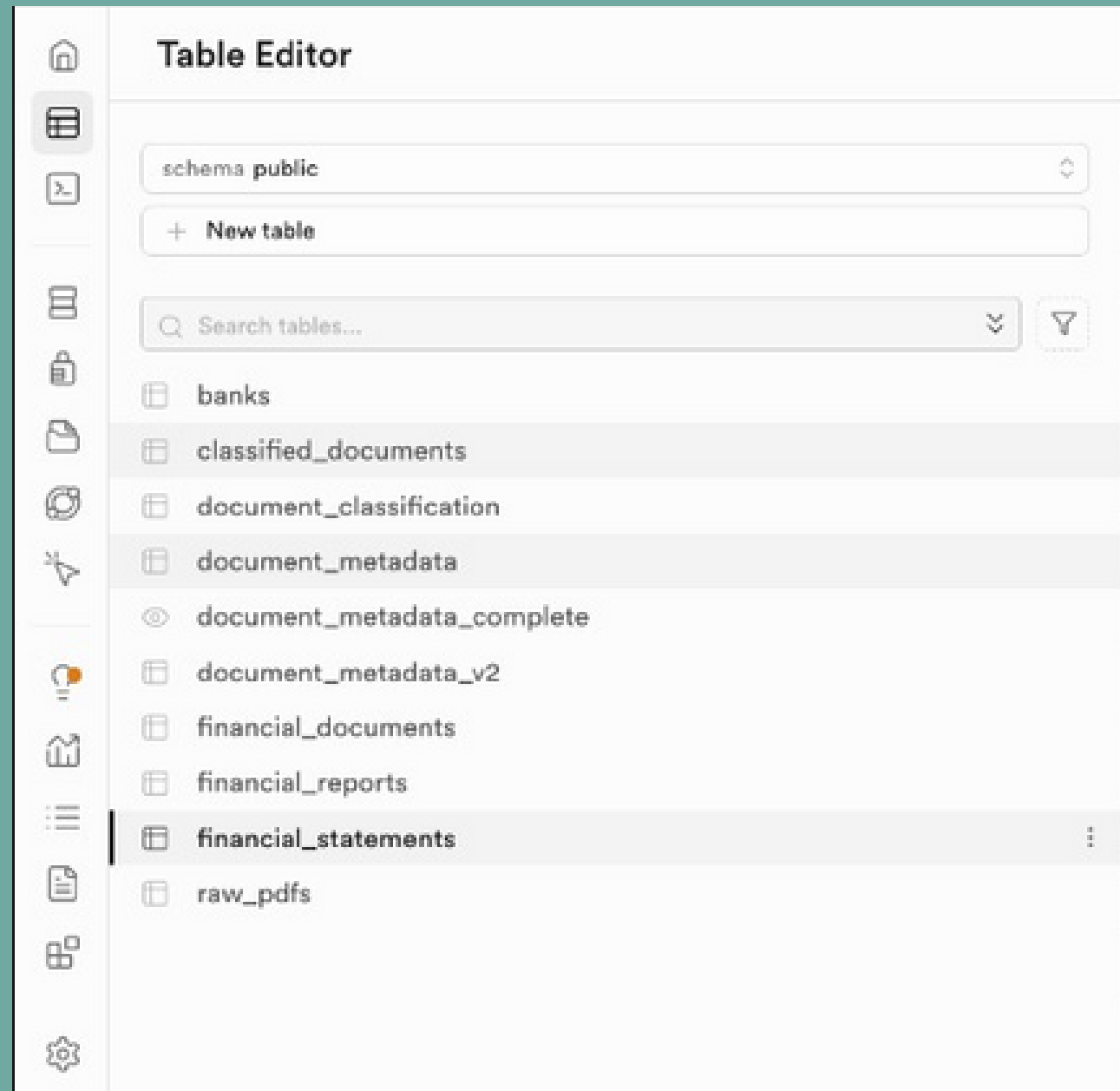


api

Provides endpoints (e.g., /get_banks, /get_financials_by_year) for backend and UI access.



Database Overview

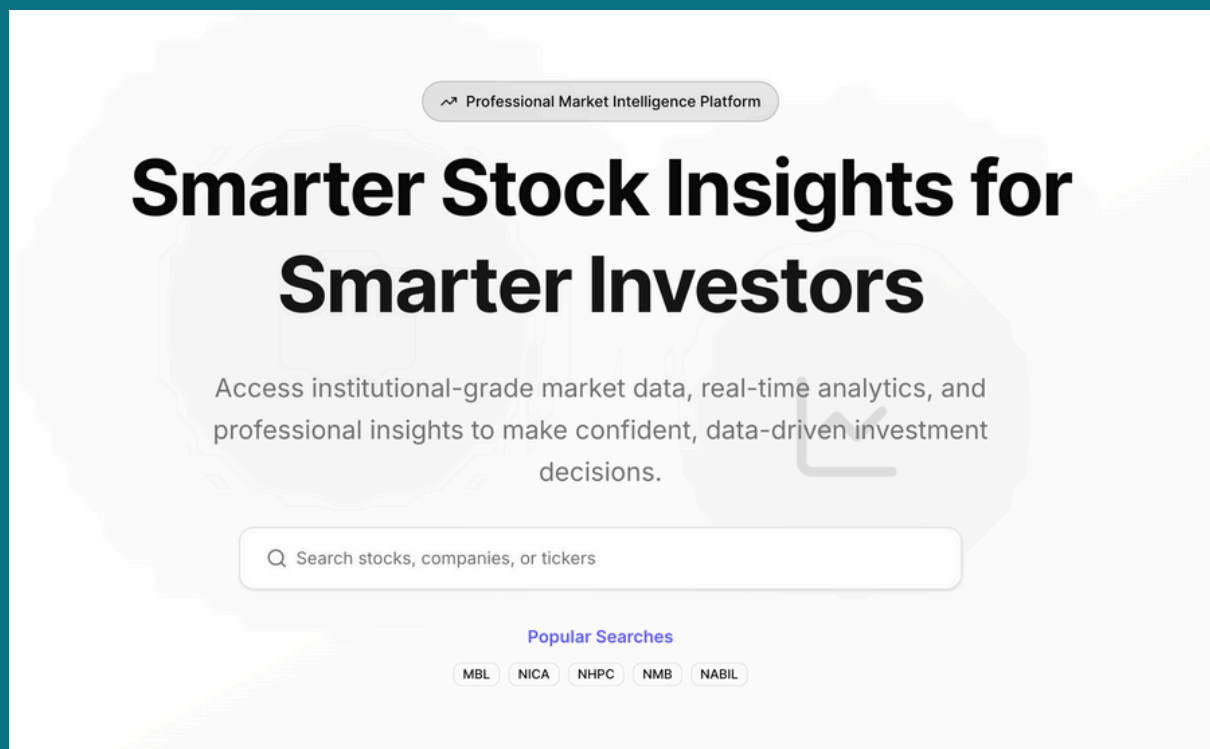


The screenshot shows the Supabase Table Editor interface for the 'banks' table. The table has 7 columns: id, symbol, bank_name, website, report_page, and annual_report_url. The data is as follows:

id	symbol	bank_name	website	report_page	annual_report_url
1	ADBL	Agriculture Development Bank Limited	https://adbl.gov.np/	NULL	https://adbl.gov.np/en/repor
2	CZBIL	Citizen Bank International Limited	https://www.ctznbank.com/	https://www.ctznbank.com/reports	NULL
3	EBL	Everest Bank Limited	https://everestbankltd.com/	NULL	https://everestbankltd.com/f
4	GBIME	Global IME Bank Limited	https://www.globalimebank.com/	NULL	https://www.globalimebank.
5	HBL	Himalayan Bank Limited	https://himalayanbank.com/	https://www.himalayanbank.com/en/publ	NULL
6	KBL	Kumari Bank Limited	https://www.kumaribank.com/	https://www.kumaribank.com/reports	NULL
7	MBL	Machhapuchchhre Bank Limited	https://www.machbank.com/	https://www.machbank.com/annual-repo	NULL
8	NABIL	Nabil Bank Limited	https://www.nabilbank.com/	https://nabilbank.com/report/investor-rel	NULL
9	NBL	Nepal Bank Limited	https://nepalbank.com.np/	https://nepalbank.com.np/reports	NULL
10	NICA	NIC ASIA Bank Limited	https://www.nicasibank.com/	https://www.nicasibank.com/investor-re	NULL
11	NMB	NMB Bank Limited	https://nmb.com.np/	NULL	https://nmb.com.np/report/s
12	PCBL	Prime Commercial Bank Ltd.	https://www.primebank.com.np/	NULL	https://www.primebank.com
13	SANIMA	Sanima Bank Limited	https://www.sanimabank.com/	https://www.sanimabank.com/know-us/r	NULL
14	SBI	Nepal SBI Bank Limited	https://nsbl.statebank/	https://nsbl.statebank/financial-reports	https://nsbl.statebank/financ
15	SBL	Siddhartha Bank Limited	https://www.siddharthabank.com/	NULL	https://www.siddharthabank
16	SCB	Standard Chartered Bank Nepal Limited	https://www.sc.com/np/	https://www.sc.com/np/investor-relation	NULL
17	PRVU	Prabhu Bank Limited	https://www.prabhubank.com/	https://www.prabhubank.com/investor-re	NULL
18	NIMB	Nepal Investment Mega Bank Limited	https://www.nimb.com.np/	https://www.nimb.com.np/digital-banking	NULL
19	LSL	Laxmi Sunrise Bank Limited	https://www.laxmisunrise.com/	https://www.laxmisunrise.com/reports/la	NULL

We used Supabase. Data stored in multiple tables. Banks, Raw URLs, Document classification, Metadata & Extracted financial data. They illustrate how financial data is organized, classified, and stored across the system.

Overview of the UI/UX



Search Functionality

- Our UI displays financial data from the PDFs of Nepali banks stored in our database
- Built for simplicity, speed, and clarity, the front-end allows any user to search, compare, and analyze companies without technical barriers.

The design follows a modern, minimalistic look, inspired by global financial terminals but adapted to Nepali market needs.

Financial Summaries

- User-friendly navigation
- Clear typography and high contrast to enhance readability
- Real-time data interaction utilizing backend FastAPI services

Financial Dashboard & Charts

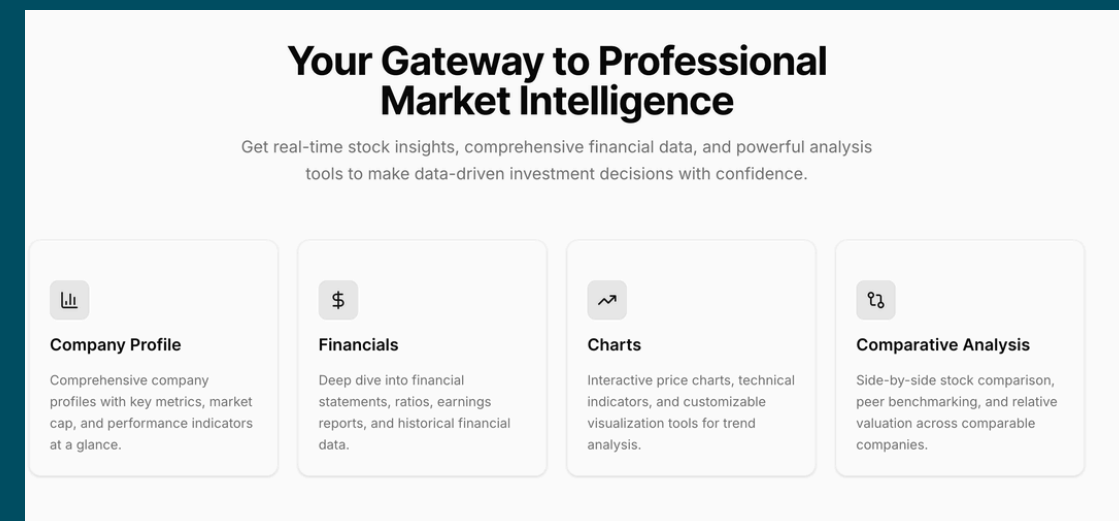
From Unstructured PDFs to Clear Financial Insights

Dashboard Features:

- **Structured, Clean Tables** for displaying the Income Statement, Balance Sheet, and Cash Flow
- **Live Year-over-Year Comparisons** sourced directly from FastAPI
- **Interactive Charts** to visualize trends such as deposits, loans, profits, and EPS
- **Automatic Mapping of Nepali fiscal years (BS)** into clear visual formats

UX Goal:

Transform complex multi-page bank PDFs into readable formats in under 10 seconds.



Search & Discovery Experience

Seamless Search Across All Nepali Companies

What the UI Offers:

- A robust search bar that allows users to enter stock symbols, company names, or categories.
- Real-time data calling from our database

Financial Highlights									
	Financials							Ratio	Valuation
Financials	2065	2066	2067	2068	2069	2070	2080	CAGR	LTM
Interest Income	1,041,473,434	1,688,617,996	2,067,746,407	1,926,128,475	2,473,773,939	2,836,363,821	16,598,693,546	48.52%	16,602,275,044
Interest Expense	580,036,192	1,144,808,132	1,544,727,637	1,500,771,881	1,486,595,730	1,720,366,986	11,610,500,770	53.43%	11,612,330,657
Net Interest Income	461,437,242	543,809,864	523,018,770	425,356,594	944,037,005	1,115,996,835	4,988,192,776	40.5%	4,989,944,387
Loan Loss Provisions	148,200,324	335,040,855	129,142,442	7,470,864	450,322,276	-10,038,632	1,398,187,605	37.8%	1,398,187,605
Non Interest Income	154,970,450	153,232,959	155,170,367	164,709,785	300,094,078	288,070,953	1,671,490,164	40.46%	1,671,490,164
Non Interest Expense	273,836,724	375,582,769	438,488,090	536,433,477	599,710,141	751,619,979	3,595,606,194	44.46%	3,595,606,194
Operating Income	468,207,368	321,460,054	110,419,204	115,983,704	793,808,807	572,186,391	4,987,980,010	40.21%	4,987,980,010
Net Income	123,251,098	73,312,799	8,923,003	38,212,959	148,599,200	454,687,791	1,045,199,187	35.72%	1,045,199,187
Eps	0	4.96	0	1.54	5.98	18.34	0	0	0
Cash Dividend Per Share	0	0	0	0	0	0	0	0	0

Why this matters:

- Nepal has no Bloomberg/Yahoo Finance-like platform
- Users usually navigate PDF archives manually

Our UI reduces the discovery time from **hours** → **seconds**

Technical Integration Behind the UI

How the UI Connects to Our Backend

- Our user interface takes raw financial PDFs from Nepali banks and converts them into a Yahoo Finance-style dashboard.
- Designed for simplicity, speed, and clarity, the front-end enables any user to search, compare, and analyze companies without facing technical hurdles.

The design embraces a modern, minimalist aesthetic, drawing inspiration from global financial terminals while being tailored to the needs of the Nepali market.

Under the Hood:

- The UI activates FastAPI endpoints, including:
 - `/financial-data/{symbol}`
 - `/financial-statistics/{symbol}`
 - `/extract-bytes` for on-demand extraction
- It utilizes structured JSON responses from Supabase.
- All heavy processing tasks (PDF parsing, Gemini extraction, classification, OCR, chunking) occur behind the scenes.
- The UI presents only clean, user-friendly outputs.

UX Philosophy:

Hide complexity. Show clarity.

Long-Term Vision

Product Vision

- Build a FactSet/Bloomberg-style financial data terminal
- Offer white-label solutions for emerging markets
- Serve universities, fintech startups, asset managers, and institutional investors

Market Expansion Roadmap

- **Phase 1:** Nepal
- **Phase 2:** Bangladesh, Pakistan, Indonesia, Thailand
- Develop backtesting + algorithmic trading tools
- Integrate LLM-powered financial analytics
- Become the leading financial data provider for frontier and emerging markets

DEMO

